

**Pre-registered report: Mapping Synesthetic Sequences in Space: Improving Consistency
Test by Harnessing Cartography Tools.**

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Abstract

Sequence-space synesthesia is a perceptual condition in which ordinal sequences (e.g., numbers, days of the week, or months of the year) are experienced as occupying specific spatial locations. This perceptual condition is identified using self-reports (e.g. questionnaires) and behavioural consistency tasks. Existing consistency tests measure the spatial distance between responses to repeated stimuli, but they ignore the ordinal and geometric features of spatial-forms. This preregistered study consists of a present and future analyses. In the present analyses, we optimise SSS diagnostics from consistency test by extracting novel geometric features at the spatial-form level. To do this, we first evaluate classification performances of old and new features in a large ($N = 685$) aggregated sample from four available and independent datasets. We then harness a geographic toolbox to extract metrics from the spatial-forms level. Finally, receiver operating characteristic (ROC) analyses are conducted to assess the discriminative performances of each method. The results show that permuted topological validity of spatial-forms and the perimeter between stimuli performs equally well in detecting SSS. In a future study, we will examine predictive validity of the selected methods on an independent dataset that has yet to be collected. Findings from the present study highlight the relevance of topological principles for sequence-space representations and aims to optimise the ability of consistency tests to detect SSS.

Keywords: Sequence space synaesthesia, Synaesthesia/synesthesia, consistency test, space, time, numbers

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Introduction

Sequence-Space Synaesthesia (SSS) or visuo-spatial forms is the phenomenon where people visualize ordered sequences in particular spatial positions. For example, numbers, weekdays or months (synesthetic *inducers*) are represented as arranged into specific spatial positions in space (synesthetic *concurrent*). The synaesthetic spatial-forms can be complex and have variable geometric forms. Those geometric forms might be different for each categories (i.e. number-forms weekdays and months), with forms such as ellipses, zig-zags or curved lines Flournoy (?). SSS are usually identified using questionnaires and consistency tests. An individual is identified as SSS based on the variability at the stimulus level between repeated responses: less variable responses being an indicator of the consistent response characteristic of SSS. However, stimulus level consistency can also be achieved by responding to the same position for all stimuli. Furthermore, it might favour linear over circular spatial-forms and is not informed about the ordinality between stimuli (e.g. $1 \rightarrow 2 \rightarrow 3$, etc.). Here, we aim at optimizing the identification of SSS from consistency tasks using geometrical characteristics of spatial-forms from ordered stimuli or inducers.

SSS frequently co-occurs with other subtypes of synaesthesia and as for other subtypes of synaesthesia it's idiosyncratic. Colour-grapheme synaesthesia, where a grapheme *inducer* triggers a *concurrent* colour (i.e. "A" seen as red), co-occurs at 71 % to 76 % with SSS (?). Nevertheless, across different synaesthetic subtypes, self-reported SSS tend to cluster together (?), indicating a degree of internal homogeneity and justifying to study it as a distinct phenomenon. Spatial-forms are idiosyncratic, which results in considerable heterogeneity in how this phenomenon is manifested across individuals. One source of heterogeneity is given by dimensionality: some SSS experiences can involve three-dimensional (3D) and two-dimensional (2D) arrangements (?; ?). Another source of heterogeneity is the reference frame, for example, the spatial-forms might take place in an external space around the body (i.e. projector) or in an egocentric internal space (i.e. associator) (?; ?). Further variability can be explained by temporal-spatial properties and mental-manipulation of spatial-forms such as

“zooming” in and out or rotating or shifting perspectives at will (?; ?) Lastly, the shape, complexity and layout of the spatial-forms are also heterogeneous such as forming for example ovals, lines, zig-zags or loops. With some recurring shapes being more frequent across SSS, such as ovals for months (?).

Despite the after mentioned heterogeneities, SSS is also phenomenological characterized. There might be five main characteristics for synaesthesia in general and specifically SSS (?; ?). *Automaticity*: the inducer automatically triggers the concurrent. *Unidirectionality*: while the *inducer* triggers the concurrent, the concurrent does not trigger the inducer. *Developmentally*: the experience was already present during childhood. *Consciousness*: The concurrent is consciously perceived. *Consistency*: the inducer-concurrent pair remains stable in time within subject. While some of these characteristics can be captured with self-reported questionnaire (i.e. for development and consciousness), other can be quantified more objectively using behavioural tasks such as consistency tests (?).

Consistency tests

Consistency tests are extracted from behavioural tasks used to evaluate how stable synesthetic experiences are over time (?; ?). They do this by presenting the same stimuli repeatedly and assessing how similar participant’s responses are across repetitions (i.e. *inducer-concurrent* associations). High consistency is often taken as evidence for synaesthesia. Consistency tests have proven effective for colour-grapheme synaesthesia. Measures of individual consistency can be derived using colour pickers while repeatedly presenting the same inducer (i.e. “A”). The quantification of the distance between concurrents from the same inducers is used to discriminate consistent synaesthetes from controls. Specifically, the euclidean distance in CIE L*u*v colour space, which is designed for perceptual uniformity, yields optimal classification accuracy when using a cut-off estimated as deduced from a larger representative sample(?). A similar rationale to that used for colour-grapheme synaesthesia has been applied to characterize SSS from consistency tasks.

In the SSS consistency task, participants complete multiple rounds of the same stimuli (e.g., digits, days of the week, and months of the year) and are asked to indicate a spatial location for each stimulus by clicking on the computer screen. Brang et al. (?) for example

evaluated consistency as the distance between repeated responses to the same inducer or stimuli (e.g. January and January) relative to adjacent stimuli (e.g. February). A response was defined as consistent if it fell within 1.96 *z-scores*. However, this criterion was noted to be potentially too conservatory, as it identified synaesthesia in 4 of 81 self-reported synaesthetes. Geometrical features such as area and perimeter between the coordinates of repeated inducers have further been used as a measure of the distance between same inducers, hence as a metric for consistency. Less consistent responses leading to smaller triangle perimeters and area (?). A cut-off corresponding to less than 0.203 % average responses area in proportion to the total screen area to classify as SSS has been suggested (see also ?).

However, several general caveats of consistency tests using concurrent or stimulus level metrics have been identified. Some tests may favour certain spatial-forms (?) in particular those with linear or highly regular spatial layouts, i.e. elliptical patterns for months(?; ?; ?). Another concern is that participants that respond with the same position, for example when not knowing the response, will have artificially high consistencies with those metrics (?). This has led to alternative approaches such as to add the standard deviation of responses, questionnaire cut-off (?) or permutation-based comparison of individual responses to chance levels for colour-grapheme(?) and SSS (?). Finally, these consistency tests evaluate the variability of responses to the same stimuli, therefore the ordinality between stimuli is not taken into account. Synaesthetic spatial-forms follow ordinal rules (e.g. Monday → Tuesday → Wednesday). In the domain of numerical cognition, ordinality has been identified as an important information when processing sequences (?). Some studies of SSS have systematically investigated ordinality by investigating metrics of adjacent inducer-concurrent pairs such as distances (?) or angles (?). However, ordinality of the concurrent spatial-form might be particularly relevant considering all the stimuli of a certain category (i.e. weekdays or months). The spatial-forms of the concurrents are often spatially structured into configuration such as lines or polygons that may follow distinctive geometrical rules. For example linear layouts have been described in early accounts of number forms (?; ?).

Present study

The goal of this registered report is to compare different consistency tests on their classification performances of individuals with SSS and controls doing the consistency task using Receiver Operator Characteristics (ROC) analyses. In the present *Phase I*, we merge four previously available datasets using the same task on both SSS and control groups (?). First, we aimed to reproduce established consistency test methods based on stimulus-level consistency metrics, such as area and perimeter. Second, we explored a new approach comparing geometrical features across repetitions of the same categories (weekday, month and numbers), such as segments and polygons. These novel features are designed to take advantage of the sequentiality or ordinality between the responses and the geometrical properties at the spatial-form level, see Figure ???. All the consistency tests are then compared using ROC analyses on their correct classification performances of self-reported SSS and control groups.

In a future study, we will assess whether the tests identified in the present study generalize and are validated on an independent dataset that is yet to be acquired (Sachdeva et al. (?), Stage 1 in-principle acceptance, see also <https://osf.io/6wn4m>).

Methods

Code and data for “one-click” reproducibility of this manuscript including the following analyses are available on git-hub (<https://github.com/remLach/SpaceSequenceSynDiagnostic>). All the analyses are conducted in R, using the following packages ‘pROC’ v. 1.19.0.1 (?), ‘sf’ v. 1.0.21 (?) and for visualization ‘ggplot2’ v. 4.0.1 (?), ‘ggVennDiagram’ v. 1.5.6 (?).

Datasets and Participants

To base the optimization of the consistency test on the largest possible sample we looked for accessible datasets that used the same consistency tasks on groups of synesthetes and controls. We found four datasets that met these criteria. Two datasets were collected in laboratory settings Rothen et al. (?) (see also <https://osf.io/6hq94>) and Van Petersen et al. (?) (see also https://data.ru.nl/collections/di/dcc/DSC_2018.00019_653). The two others came from on-line testing Ward (?) (see also <https://osf.io/nu5v4>) and an additional dataset was gently provided by private communications with Professor Ward. To match the other datasets,

stimuli of the weekdays and months categories were translated from Dutch to English in the dataset from (?). For the number category we included only digits from 0 to 9 that were presented across all datasets (i.e. excluding the stimuli “50” and “100” for numbers in Van Petersen et al. (?)).

From the merged datasets, we kept 685 from the total 689 participants. First, we excluded 0.02 % empty trials (i.e. skipped responses) including trials flagged for having the same x or y coordinates across conditions and repetitions, causing the depletion $n = 2$ participants (as in ?; ?). $n = 2$ participants were excluded for having less than 4 coordinate points since polygons need at least four coordinate. Note therefore, that 31 participants of the final sample did not have responses in all three conditions X repetition cases. From the final sample of $N = 685$, 396 were synaesthetes and 289 controls, see Table ?? and Table ??.

Regarding the synaesthetes profiles, detailed participant information such as questionnaire responses were only available from the two datasets from the data by Professor Ward (i.e. a majority of 573 participants). We described the self-reported profiles for the stimulus categories used in the consistency test (i.e. number, weekdays and month), see Figure ??. From this Venn-diagram we see that 240 of the SSS report having spatial-forms for Numbers, Days and Month; 42 only Days and Month; 21 only Number and Day. Also, 127 controls report spatial-forms for either Days, Numbers, Months or combinations.

Procedure

For the consistency task, each stimuli were presented randomly and sequentially centrally on the screen. Participant were instructed to click on the screen position where they visualize them. In Van Petersen et al. (?) and Rothen et al. (?) the participants were allowed to skip responses.

Stimulus included here were 7 weekdays (Monday to Sunday), 12 months (January to December) and 9 numbers (0 to 9). For all the data from Professor Ward the stimuli were presented in randomized order with the constraint that no stimulus was repeated until all unique stimuli ($N = 29$) had been presented once. The median display resolution was 1440 X 768, with a maximum of 2560 X 2025 and a minimum of 308 X 149.

Analyses

Consistency tests on the stimulus level

We reproduced the tests extracted from consistency tasks found in the literature such as area and perimeter between repetitions (?: ?; ?).

Area and perimeter between repetitions. Formally, the area (see Equation ??) and perimeter (see Equation ??) are calculated from the triangle (i.e. given three repetitions) formed by the x and y coordinates of the three repetitions of each stimuli (i.e. $(x1, y1)$, $(x2, y2)$, $(x3, y3)$).

$$Area = \frac{|x1y2 + x2y3 + x3y1 - x1y3 - x2y1 - x3y2|}{2} \quad (1)$$

$$Perimeter = \sqrt{(x2 - x1)^2 + (y2 - y1)^2} + \sqrt{(x3 - x2)^2 + (y3 - y2)^2} + \sqrt{(x1 - x3)^2 + (y1 - y3)^2} \quad (2)$$

Each stimuli's area was then averaged by participants. The area metric is in % of the screen area to be able to compare with the consistency across studies using different screen sizes. To account for screen size and spread variability across the datasets and participants, area and perimeter were also calculated on individually normalized x and y coordinates (z -score). These methods compute consistency metrics at the stimulus level, the rationale is that consistent responses would show less variability between repetitions. For example, smaller area or perimeter of the triangle formed indicate more variability hence more consistent responses between the three repetitions.

Consistency tests on the space-form level

For the novel consistency tests, we extracted features at the spatial-form level for each repetition separately. Taking the stimuli as an ordered sequence we can consider them as a geometrical segments (i.e. open geometrical form) and polygons (i.e. closed geometrical form), similarly as originally described in Galton (?).

We harnessed a geospatial analysis, the 'sf' package version 1.0.21 (?) to extract polyons and geometry-based features from participant's 2D (x,y) coordinate responses. This

package allows, for example, to build and analyse segments or polygons and then extract multiple geometrical descriptors or features. Informed by the ordinality of the stimulus, we defined segments and polygon by conditions and repetitions. The rationale here was to determine whether, when considering the spatial-forms formed by stimuli as ordered coordinates (i.e. as segments or polygon) they remain consistent *between* repetitions for each individuals. We extracted the topological validity, topological simplicity and a compactness score for each polygons (i.e. 3 conditions X 3 repetitions: 9 polygons per participants).

Segment self-intersections. The first consistency test extracted from the segments is self-intersection. We considered the spatial-forms as a segment constituted by the lines connecting consecutive stimuli separately for each category and repetitions (i.e. Monday (repetition 1) → Tuesday (repetition 1) → Wednesday (repetition 1), ect) and then counted the number of times each segment self-intersect. The rationale is that SSS should have less chance to produce that self-intersect than control. The number of self-intersections were computed separately for each repetitions and conditions and averaged per participants.

Topological validity and simplicity. We assessed topological validity using geometric validation test. Topological validity tests if a polygons is well-formed and valid according to the Open Geospatial Consortium (OGC) Simple Features Specification (?). A polygon is considered topologically valid if it satisfies the following criteria: (1) if polygon rings are simple (i.e., they do not touch or self-intersect), (2) boundary rings to not cross (3) boundary rings may only touch points tangentially (4) rings that define holes are contained within the exterior ring (5) the polygon rings must not splits the polygon (?).

Topological simplicity evaluates whether a geometric object has a simple structure without self-intersections or self-tangencies. According to the OGC Simple Features Specification (?), a polygon boundary is considered simple if it does not pass through the same point more than once (?). For polygons, simplicity requires that each ring does not self-intersect or self-touch. Note that simplicity is a condition for validity: a polygon can be simple but still invalid (e.g., if an interior ring extends outside the exterior ring).

For each participant we tested for topological validity and simplicity across individual 3 categories (weekdays, month and numbers) and 3 repetitions, hence 9 forms. The binary

outcome (1 = is valid, 0 = invalid) was then averaged across all 9 forms. Hence, it can span from 1 (all 9 are valid/simple) to 0 (none of the 9 forms are valid/simple). Thus, the topological validity and simplicity score ranges from 0 (none of the nine polygons are valid/simple) to 1 (all nine polygons are valid). For example, if a participant's polygon for weekdays were valid across all three repetitions, months were valid in two of three repetitions, and numbers were invalid in all repetitions, the topological validity score would be $5/9 \approx 0.56$.

Compactness score. We also implemented a measure of compactness for the polygons. We used the score developed to quantify the gerrymandering of political district by Polsby and Popper (?). Gerrymandering is a politically motivated strategy of redefining district's boundary to give the advantage to a political party. This has the consequences to reduce the compactness of those districts as would be expected if they were defined from geomorphology. The compactness is calculated as the ratio of the area and perimeter of a polygon. The rationale is that more compact forms (i.e. higher Polsby–Popper scores) would be more likely in SSS than controls. Formally, the Polsby–Popper score is calculated from the polygon's area and perimeter Equation ??:

$$score = \frac{4\pi \cdot Area}{Perimeter^2} \quad (3)$$

Permutation-based feature extraction. Finally, we implemented permutations of the repetition order to derive a permuted measure of consistency. Specifically, instead of using repetitions in chronological order, we shuffled the repetition order between stimulus presentation (i.e. Monday (repetition 2) → Tuesday (repetition 3) → Wednesday (repetition 1), ect) and extracted the geometrical feature from x,y coordinates of same stimulus categories in non-chronological order. The form-based features computed before were relying on the chronologically ordered repetitions. For example, when a stimulus such as Monday was repeated three times, the coordinates for the first presentation of Monday were always paired with the coordinates for the first presentation of Tuesday to construct segments or polygons. However, if synesthetic forms are truly consistent, they should remain stable independently of the chronological order of stimulus presentation. Hence, we permute the repetitions within each condition. Specifically, for each participant and each category, we randomized the

response repetition order. For example segments for weekdays were constructed with the x,y coordinates where randomly permuted across repetition order, for example Monday (repetition 1), Tuesday (repetition 3), Wednesday (repetition 2), ect. We predicted that this permuted-based consistency would yield to better classification performance (higher AUC values) compared to chronologically ordered features. The rationale is that genuine synaesthetes would have consistent topological features of spatial-forms repetitions regardless of presentation order, whereas controls would show random variation. The permutation scores were calculated for topological validity and Polsby–Popper score.

Receiver Operator Characteristic analyses

Each of the previous method's performance in classifying SSS from controls was compared with Receiver Operator Characteristics (ROC) analyses. Classification performance was quantified using the area under the curve (AUC), with higher AUC values indicating better discrimination between SSS and control groups. The optimal cutoffs were calculated using Youden's index(?). Discriminant Power (DP) was used as an additional estimate of discrimination performance according to the optimal cutoff (Equation ??). DP around 1 being interpreted as inefficient discrimination.

$$DP = \frac{\sqrt{3}}{\pi} (\log(\frac{sensitivity}{1sensitivity}) + \log(\frac{specificity}{1specificity})) \quad (4)$$

The statistical comparison of ROC curves between the methods was operated testing the difference of AUC between pairs of curves using bootstrapping method (1000 times) since it allows to compare methods with different directions. Each bootstrap results in the subtraction of the AUC between the two ROC. Then the each of the AUC difference's (D) are divided by the standard deviation of those difference and compared to a normal distribution (Equation ??).

$$D = \frac{AUC_1 - AUC_2}{SD} \quad (5)$$

Therefore we statistically compared all the AUC of each methods as the difference to the highest AUC method.

Results

Descriptively, the triangle area (in % screen size) between repetition for SSS was surprisingly larger than in previous studies: 0.26% ($SD = 0.52\%$) compared to 0.15% ($SD = 0.13\%$) in Rothen et al. (?) and 0.14 % ($SD = 0.17\%$) % in Ward et al. (?), see Table ???. These values also differ between the available datasets (See Figure ??). These differences could be mitigated by using individually standardized coordinates $M = 0.05$ ($SD = 0.06$) Rothen et al. (?), $M = 0.08$ ($SD = 0.15$) Ward (?), $M = 0.09$ ($SD = 0.15$), Ward2, see also Table ??.

ROC analyses

All ROC were compared to the permuted validity method, since it led the highest AUC (81.16), all p-values false discovery rate (FDR) adjusted. ROC analyses are summarized in Table ?? and Figure ?. Interestingly, while the permuted validity leads to significantly larger AUC than the Polsby-Popper method, ($D = 2.14$, $p < .05$), it led to similar AUC than the standardized perimeter ($D = 1.4$, $p < .05$). This might be explained by the relatively lower sensitivity 59.7 (although relatively higher sensitivity 87.54) of the Polsby-Popper test compared to the Perimeter (72.54) and permuted validity tests (70.78). Additionally, the permuted validity led to significantly higher AUC than the non permuted one ($D = 3.12$, $p < .05$). Both criteria display bimodal distributions across the groups (see Figure ??).

Hence, the two best AUC performances are from the averaged permuted validity score (AUC = 81.16; DP = 2.13, cut-off = 0.17 corresponding to 1.53 valid of 9 forms) and the other for standardized perimeter between the repetitions of each stimuli (AUC = 78.8; DP = 2.06, cut-off = 1.73 z-scores). However, at the proposed cut-off, the permuted validity leads to higher specificity than perimeter (78.55 vs. 75.78) indicating better rejection of controls (i.e. less false negatives) but lower sensitivity (70.78 vs. 72.54), indicating poorer detection of SSS (i.e. more false negatives). See Figure ?? for the densities of each standardized features suggesting bi-modal distribution.

In sum, the permuted validity and perimeter both provide similarly good AUC compared to the other consistency tests.

Further analyses

We conducted further analyses by subsampling them based on the dataset source, questionnaire percentiles. The aim of these analyses was to test the stability of the classifications from different methods across experiments (i.e. datasets) and sample characteristics (i.e. questionnaire scores). We also used a correlational approach to test for correlations between features and questionnaires.

We sub-sampled the data based on the source of the data and found different features best fit different datasets, see Figure ???. Interestingly the best features by datasets differ between the data sources. While the standardized area indeed lead to the best AUC for the data from Rothen et al. (?), the perimeter leads to better AUC in Ward et al. (?). The permuted validity leads to $AUC > 80$ across all datasets.

To test the stability of AUC of all features, we computed them across several percentiles based on the questionnaire scores. Again, only the data from Professor Ward is included there. We computed AUC, sensitivity and sensibility on sub samples based on the percentiles of the questionnaire scores (see ?), taking the 10-0% highest questionnaire scores against the 10 % worst (i.e. 90-100 %) . We further sub-sampled by 10 % questionnaire scores and recalculated AUC, sensitivity and sensibility, see Figure ??? for AUC, DP, sensitivity and specificity. Using this method we can see that the total AUC remains relatively stable across percentiles.

Another important point addressed in the literature about consistency test is circularity. The circularity is given in that if consistency tests are used to classify SSS and controls, then those groups will by definition differ in consistency (?). Hence, we made a correlation matrix with the questionnaire scores and all the features, see Appendix. The two highest correlations are indeed between the questionnaire score and perimeter ($r = .58$) and permuted validity ($r = .50$), see Figure ???.

Some of the participants might have been originally mis-attributed to control or SSS, see Figure ???. Descriptively, we also found about ~6 % of the full sample's synesthetes to be consistently classified as controls (i.e. consistent false positive) by the five methods with the highest AUC Figure ???. ~3 % controls are consistently classified as SSS (i.e. consistent false negative) Figure ???.

We also wanted to visualize the average space form of each category at the population level to see if there were qualitative differences between the forms. The figure in the appendix (Figure ??), shows more circular patterns for months and more linear horizontal and vertical for numbers.

Methods - future study

Additional data will be collected in the future using the same consistency test, with the procedural exception that there will be four repetitions per stimuli instead of three (Sachdeva et al. (?), Stage 1 in-principle acceptance). Hence for the stimulus levels feature we will compare the area and perimeter of a quadrilateral of four coordinates pairs instead of a triangle. Materials are more details on this study are pre-registered on OSF, including ample size and recruitment method (?, Stage 1 in-principle acceptance) (<https://osf.io/6wn4m>).

We will compute the same ROC analyses as in this pre-registration on all the features. Since we obtain different best AUC per features depending on the dataset (see Figure ??) it is likely that the best features here will not match with the best feature in the to be collected dataset.

Discussion - present study

Our investigation of four datasets found two main consistency tests leading to the optimal classification of SSS from control: standardized perimeter on the stimulus level and permuted topological validity on the space-form level. Hence, the area test from Rothen et al. (?) led to worse classification than the perimeter test with a larger sample. Furthermore we could prove similar performances using a different approach based on the space-form level (i.e. topological validity).

Further analyses indicates variable optimal tests in terms of AUC depending on the datasets (see Figure ??). Considering only the data by Professor Ward that contains questionnaire data, we comparable patterns of AUC and DP for more extreme Synaesthete and Control groups when sub-sampled by questionnaire's score distribution percentiles (see Figure ??). Nevertheless the standardized perimeter seems to lead to better AUC and DP in more extreme scores. Also, the standardized perimeter lead to slightly higher correlation with the questionnaire scores ($r = .57$) than permuted topological validity ($r = .51$), see Figure ??.

Limitations

Although an optimal test to classify SSS might be particularly relevant for experimental purposes, several important limitations need to be considered. First, consistency tests contain only a limited set of sequential stimuli (i.e. months, weeks and the first ten natural numbers) that could potentially elicit synaesthetic experiences. Other ordinal categories such as temperature, clock time, musical keys might be more relevant for some individuals with SSS. For numbers specifically, better consistency tests might be obtained using a larger set size (i.e. including decades and hundreds), as descriptively interesting form changes occur at different decimals in base-10 number systems (?).

Second, the use of diagnostic cutoffs assumes categorical distinctions between groups, but SSS may exist on a continuum and scores might be more suitable (?). This limitation is related to the circularity mentioned previously: diagnostic cutoffs as calculated here depend on how SSS and controls are classified in the first place (?). Measures of other characteristics of synaesthesia such as automaticity or visual Gabor detection might be necessary to establish external validity (?). Indeed, determining the prevalence of SSS in the general population requires definitional choices that might be based on more conservative or lenient criteria (??; ??; ??; ?). These difficulties are reflected in the span of current prevalence estimates of SSS: 4.4 % (?), 8.1 % (?) and 14 % (?).

Third, as shown in Figure ??, we find that the optimal criteria can vary across datasets. In other word, we obtain different AUC depending on the datasets. These differences might be explained by different sampling methods biases, for example (?) recruited SSS from over one hundred candidates to maximise participant reporting synesthetic experiences.

Topological validity

Surprisingly, permuted topological validity of the spatial-forms led to similar results than the perimeter (or the distance) between the responses. This result might be informative about how SSS map ordinal stimuli in space. Indeed, it seems the patterns of spatial-forms follow topological rules analogous to geographical and geometry-based space structures. Analogies between maps and neuroscience have a long history (i.e. retinotopy, sonotopy or somatotopy ?). Therefore spatial-forms might originate from the mapping between ordinal or

sequential stimuli on idiosyncratic visuo-spatial abilities following topological rules. One of the advantages of using topological validity is that it also classifies responses with the same coordinates as invalid and hence inconsistent, while using the area and perimeter metrics they would qualify as highly consistent responses.

Intermediary Conclusion

The present study compared traditional stimulus-level consistency tests with novel form-level geometric tests for detecting SSS. We found optimal classification performance from permuted topological validity and standardized perimeter between repetitions. Despite not yielding to better classification than stimulus level methods, this new method is theoretically meaningful since genuine synesthetes should produce more compact and topologically valid structures compared to controls. Similarly, the perimeter test captures the stability of the overall spatial configuration across repetitions, which should remain consistent for true SSS individuals regardless of presentation order.

In the future study, we will attempt at validating these criteria on a yet-to be acquired dataset.

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Appendix

The additional analyses in this appendix attempt to address several concerns. Regarding the distribution's of the scores from the different methods, i.e. whether it is bimodal, see Appendix A. Then to visualize the stability of the ROC analyses across the datasets, i.e. whether AUC, DP, sensitivity and specificity vary across the four different datasets, see Appendix B.

To address circularity concerns that we might find methods that best classifies synesthetes and controls only from the sed on self-reported groups, we attempted two additional approaches: first, testing sub-samples based on percentiles of the questionnaire score sample distribution (see Appendix C). The rationale is that a good consistency test should have similar discriminative performance for strong SSS and weaker SSS. To do this, we re-sampled the SSS and controls based on their questionnaire distribution by percentiles (i.e., comparing 10% highest questionnaire scores vs. 10% lowest, then 20% vs. 20%, etc.). In addition, we correlated the questionnaire scores with the results from different features extracted from the consistency test, see Appendix D.

We also wanted to visually characterize whether certain patterns occur more frequently for some categories than others (i.e. circular pattern for months and linear for numbers). Appendix E shows the average forms for each categories. Finally, we wanted to visualize how each participant is classified by each test. From this we found a group of Synaesthetic who consistently are classified as controls by different methods as well as a group of control who are consistently classified as synaesthetes, see Appendix F.

The following R packages were used for the data visualization in the appendix:
'ggribes' v. 0.5.7 (?), 'cowplot' v. 1.2.0 (?), 'ggalluvial' v. 0.12.5 (?), 'corrplot' v. 0.95 (?)

Table 1*Synesthetes*

Synaesthetes	N before	N after	Age
(?)	33	32	23.1
(?)	23	22	23.2
(?)	252	252	37.2
Ward 2	90	90	NA
Total:	398	396	

Note. Synesthetes. Demographics were not provided in the dataset Ward 2. Before: before participant exclusion

Table 2*Controls*

Controls	N before	N after	Age
(?)	37	37	28.2
(?)	21	21	21.6
(?)	215	213	19.9
Ward 2	18	18	NA
Total	291	289	

Note. Controls. Demographics were not provided in the dataset Ward 2. Before: before participant exclusion

Table 3*ROC analyses of all features*

	Feature	AUC	DP	threshold	sensitivity	specificity
10	Permuted valid structure [boolean/9]	81.16	2.13	0.17	70.78	78.55
11	Polsby–Popper [score/9]	78.89	2.29	0.19	59.70	87.54
4	Perimeter [z-score]	78.80	2.06	1.73	72.54	75.78
9	Valid structure [boolean/9]	78.12	1.90	0.14	72.80	72.32
12	Permuted P-P [score/9]	77.15	2.46	0.20	57.18	90.31
5	Segment self-intersections [n/9]	72.93	1.75	1.17	79.85	60.21
6	Area of the polygon [z-score]	72.03	1.36	1.29	64.99	68.51
8	Simple topology [boolean/9]	69.74	1.24	0.28	61.96	68.51
2	Area [z-score]	69.27	1.68	0.08	76.32	63.32
7	Perimeter of the polygon [z-score]	59.12	1.29	10.46	83.88	41.87
3	Perimeter [screen size %]	55.34	0.68	0.03	76.07	38.75
1	Area [screen size %]	50.87	0.79	0.21	76.83	40.48

Table 4*Descriptives of each features*

Feature	Control	SSS	Control [zs]	SSS [zs]
Permuted valid structure [boolean/9]	0.12 (0.16)	0.36 (0.25)	-0.57 (0.64)	0.41 (1.01)
Polsby–Popper [score/9]	0.11 (0.13)	0.27 (0.19)	-0.51 (0.69)	0.37 (1.03)
Perimeter [z-score]	3.30 (1.72)	1.59 (1.16)	0.60 (1.04)	-0.44 (0.70)
Valid structure [boolean/9]	0.13 (0.19)	0.39 (0.28)	-0.54 (0.69)	0.39 (1.01)
Permuted P-P [score/9]	0.11 (0.17)	0.41 (0.33)	-0.56 (0.56)	0.40 (1.06)
Segment self-intersections [n/9]	9.56 (11.31)	1.87 (5.42)	0.48 (1.23)	-0.35 (0.59)
Area of the polygon [z-score]	1.00 (0.99)	1.91 (1.31)	-0.42 (0.79)	0.30 (1.03)
Simple topology [boolean/9]	0.22 (0.23)	0.40 (0.27)	-0.39 (0.85)	0.28 (1.01)
Area [z-score]	0.26 (0.32)	0.08 (0.14)	0.42 (1.29)	-0.30 (0.55)
Perimeter of the polygon [z-score]	9.49 (3.44)	8.39 (2.48)	0.21 (1.16)	-0.16 (0.83)
Perimeter [screen size %]	0.03 (0.03)	0.03 (0.08)	0.03 (0.44)	-0.02 (1.26)
Area [screen size %]	0.46 (0.87)	0.26 (0.52)	0.17 (1.25)	-0.12 (0.75)